

A2: TEC Heating and Cooling Analysis

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Steady temperature vs signed PWM

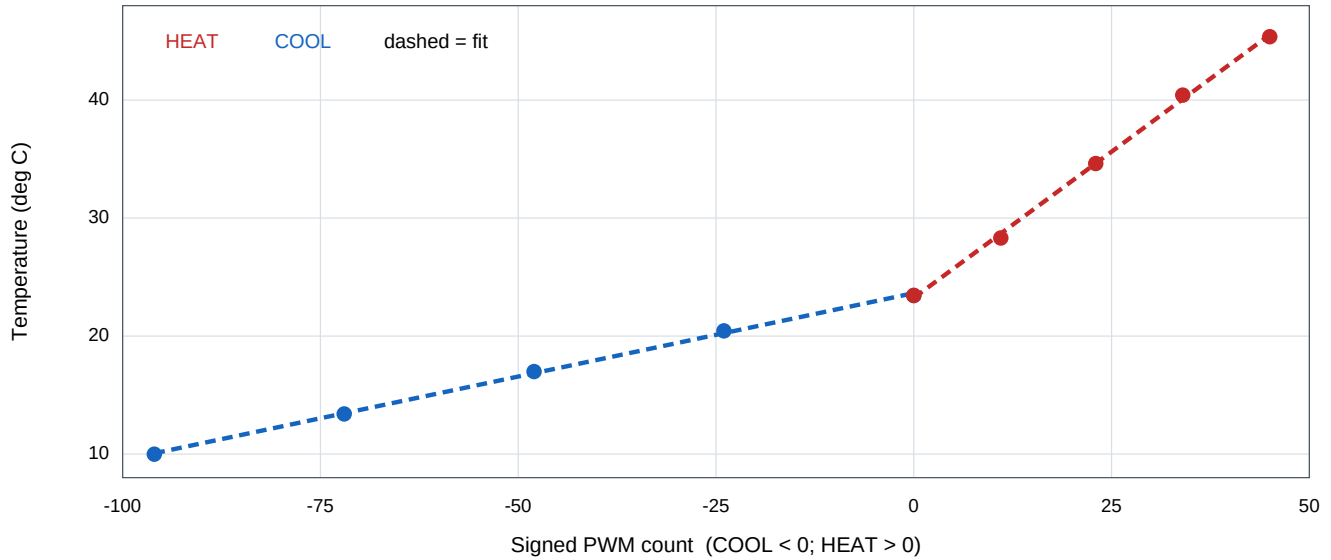


Figure 1. Five selected PWM magnitudes per direction; the two 0-PWM points overlap at 23.44 deg C. Steady temperatures are the means of the final approximately 20 s after the trace ceased sustained directional drift and fluctuated within a local band. Fits cover HEAT 0-45 and COOL -96-0 signed PWM counts.

HEAT slope	COOL slope	Measured ratio	Inferred Joule/Peltier
0.4954 deg C/count	0.1414 deg C/count	3.5035	0.5559

PWM averaging and steady-state energy balance

During one PWM period, the signed on-state current flows for fraction D of the period and is zero otherwise. Here D is the magnitude of signed PWM divided by 255. Evaluating the two period integrals gives:

$$\langle I \rangle = (1/\tau) \int_0^\tau I(t) dt = [I(D\tau) + 0(1-D)\tau]/\tau = DI \quad \langle I^2 \rangle = (1/\tau) \int_0^\tau I(t)^2 dt = [I^2(D\tau)]/\tau = DI^2$$

$$\langle I^2 \rangle - \langle I \rangle^2 = D(1-D)I^2$$

The last expression is nonzero for duty cycles strictly between zero and one. Using the square of the mean current instead of the mean squared current would incorrectly make Joule heating quadratic in duty. The measured branches are approximately linear.

Let signed duty equal signed PWM divided by 255. The reduced heat balance counts TEC conduction and other passive leaks once in the effective conductance:

$$C dT/dt = Q_{TEC} - G(T-T_0); \quad Q_{TEC} = dQ_P + |d|Q_J \quad dT/dt = 0: \quad G(T-T_0) = Q_{TEC}$$

The Peltier term reverses between heating and cooling; the Joule term does not. Solving both branches and differentiating with respect to signed PWM gives:

$$T_h - T_0 = d(Q_P + Q_J)/G; \quad T_c - T_0 = d(Q_P - Q_J)/G \quad m_h = (Q_P + Q_J)/(255G); \quad m_c = (Q_P - Q_J)/(255G)$$

$$r = m_h/m_c; \quad Q_J/Q_P = (r-1)/(r+1) = 0.5559$$

Manufacturer comparison and interpretation

Laird CP14-127-045 | hot-side temperature 27 deg C

Manufacturer specifications

Values located in the 27 deg C hot-side table of the Laird CP14-127-045 data sheet. The condition and meaning of each value are shown below.

Specification	Meaning and operating condition	Value
R_M	Module electrical resistance at 27 deg C hot side	1.50 ohm
$I_{\max}$	Maximum specified current (I at Delta T _{max}) at 27 deg C hot side	8.6 A
$Q_{c,\max}$	Maximum cold-side pumping at Delta T = 0	71.3 W
$\Delta T_{\max}$	Maximum no-load face temperature difference (at Q _c = 0)	70.5 deg C

Source: Laird, *CP14-127-045 Thermoelectric Cooler*, 27 deg C hot-side specifications:
<https://sethfraden.github.io/Phys39F26-course/references/laird-tec-cp14-127-045.pdf>

Maximum-current calculation

At zero face-temperature difference, passive TEC conduction vanishes. The symmetric model assigns half the total Joule heat to the object face:

$$Q_{J,\max} = (1/2)(I_{\max})^2 R_M = 55.47 \text{ W} \qquad Q_{P,\max} = Q_{c,\max} + Q_{J,\max} = 126.77 \text{ W}$$
$$r_{\text{Laird},\max} = 1 + (I_{\max})^2 R_M / Q_{c,\max} = 2.556$$

Comparison

The measured ratio is 3.5035 versus the manufacturer's maximum-current prediction of 2.556, about 1.37 times larger. Full PWM duty means the H-bridge is continuously on, not that the current equals $I_{\max}$; the actual current depends on the supply settings (12 V, 10 A limit), H-bridge and wiring drops, and module resistance. Because the symmetric model predicts r approaching 1 below $I_{\max}$, the larger measured ratio reflects its idealizations (equal half-Joule split, single symmetric conductance) plus PWM versus DC, finite face temperature difference, changing properties, and one-slope fits over slightly curved data.

Passive conduction

When the block is hotter than room temperature, passive heat flows out; when colder, it flows in. The passive-conductance term therefore opposes both excursions. If that conductance is approximately symmetric, it reduces both responses but does not by itself explain why the HEAT slope magnitude exceeds the COOL slope magnitude. Reversing current reverses Peltier transport, whereas Joule heating keeps the same sign.

Conclusion

Our open-loop measurements show an approximately linear steady-temperature response to PWM in each direction, with a larger heating susceptibility than cooling susceptibility. The fitted slope ratio is 3.50. Under the simplified near-room-temperature energy balance, that ratio corresponds to object-face Joule heating about 0.56 times the full-on Peltier heat rate. This is a model-based inference, not a direct measurement of either heat flow. Peltier transport reverses with current, while Joule heating keeps the same sign, so their effects add during heating and partly offset during cooling. Passive conduction carries heat away from a hot block and toward a cold one, opposing both departures. The measured slope ratio exceeds the manufacturer's maximum-current prediction (2.56), reflecting the symmetric model's simplifications.